

CO4 – AT1

Coding Challenge Task

SOURCE CODE:

Q2. Unbounded Knapsack

```
#include <stdio.h>

int main() {

    int n, W;

    printf("Enter number of items: ");

    scanf("%d", &n);

    int value[n], weight[n];

    printf("Enter values: ");

    for (int i = 0; i < n; i++)

        scanf("%d", &value[i]);

    printf("Enter weights: ");

    for (int i = 0; i < n; i++)

        scanf("%d", &weight[i]);

    printf("Enter capacity: ");

    scanf("%d", &W);

    int dp[W + 1];

    for (int w = 0; w <= W; w++)

        dp[w] = 0;
```

```
for (int w = 0; w <= W; w++) {  
    for (int i = 0; i < n; i++) {  
        if (weight[i] <= w) {  
            if (dp[w - weight[i]] + value[i] > dp[w])  
                dp[w] = dp[w - weight[i]] + value[i];  
        }  
    }  
}  
  
printf("Maximum Profit = %d\n", dp[W]);  
  
return 0;  
}
```